Lesson plan · for the teacher

# Break it down

Programming · Level 1 · Lesson 4

**Time**
40 min

**Materials**
Board and chalk; Paper for maps and algorithms; Your phone (optional) with this lesson downloaded

## Learning goals

- Name the four parts of computational thinking.
- Compare two algorithms by counting steps.
- Plan a computational-thinking moment in your subject.

## Activity: Guess my number

1. Warm-up (5 min): Think of a number from 1 to 100. Pupils guess; you say “higher” or “lower”. Tally the guesses on the board.
2. Teach (10 min): Name the four parts: decomposition, patterns, abstraction, algorithms. Give one school example of each and ask pupils for another.
3. Halving (10 min): Teach “guess the middle, cross out half”. Play again in pairs and compare the counts with the warm-up. Ask why halving needs the numbers in order.
4. Plan (10 min): Groups take a task from another subject (a recipe, a journey, an experiment) and find the four parts in it.
5. Wrap-up (5 min): Ask which part was hardest to spot, and where else halving helps. Then give the exit ticket.

**Keep the board tally of guesses: the drop from the warm-up to halving is the lesson’s best evidence.**

## Discussion

- Why does halving need the numbers in order?
- Which part of computational thinking do you already teach without naming it?
- Where else could we compare two ways of doing a job by counting steps?

## Exit ticket

- Name the four parts, with one example of each from today.

**Try it in class tomorrow**
Play “guess my number” from 1 to 100. Let pupils guess freely and count the guesses, then teach halving. Can anyone beat 7 guesses?

Worksheet · for pupils

# Break it down

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Name _____ Class _____ Date _____

**1** **Which of these is decomposition?**

- ◯ Splitting a class play into scenes, costumes and music
- ◯ A seating plan that shows only the names
- ◯ Noticing that every song has a chorus

**2** **Match each part of computational thinking to a school example.**

| | | |
|---|---|---|
| Decomposition | • • • • | The steps of long division |
| Pattern recognition | • • • • | A bus map that shows only the stops |
| Abstraction | • • • • | Noticing that every multiple of 5 ends in 0 or 5 |
| Algorithm | • • • • | Splitting a science project into tasks for each group |

**3** **This short program makes the first guess in "guess my number" from 1 to 100. What does it print? (// divides and drops the remainder.)**

```
low = 1
high = 100
guess = (low + high) // 2
print("Is it", guess)
```

- ◯ Is it 50
- ◯ Is it 100
- ◯ Is it 1

**4** **Put the steps of the halving strategy in order.**

- ☐ Repeat with the half that is left
- ☐ Cross out the half it can't be in
- ☐ Hear "higher" or "lower"
- ☐ Guess the middle number

**5** **With halving, about how many guesses can it take to find a number from 1 to 1,000?**

- ◯ About 10
- ◯ About 100
- ◯ About 500

**In which topic you teach next week could pupils use decomposition, patterns or abstraction? Write the moment, in two or three sentences.**